BCA-201: Mathematics-II (Calculus)

Weightage: 70 Marks

Q1. Explain the limit of a function.

The limit of a function f(x) at x = a is the value that f(x) approaches as x gets closer and closer to a. For the limit to exist, both left-hand and right-hand limits must be equal.

$$\lim_{x \to a} f(x) = L$$

Q2. Define Continuity of a function at a point.

A function f(x) is continuous at x = a if the limit exists at that point, the function is defined at that point, and the limit equals the function value.

$$\lim_{x \to a} f(x) = f(a)$$

Q3. What is Differentiability?

Differentiability refers to the existence of a derivative at a point. If a function is differentiable at a point, it must also be continuous there.

$$f'(a) = \lim_{h \to 0} [f(a+h) - f(a)] / h$$

Q4. State Rolle's Theorem.

If f is continuous on [a, b], differentiable on (a, b), and f(a) = f(b), then there exists at least one c in (a, b) such that f'(c) = 0.

$$f'(c) = 0$$

Q5. State Lagrange's Mean Value Theorem (LMVT).

If f is continuous on [a, b] and differentiable on (a, b), then there is at least one c in (a, b) such that the tangent is parallel to the secant line.

$$f'(c) = [f(b) - f(a)] / (b - a)$$

Q6. What are Taylor's and Maclaurin's Theorems?

These theorems express a function as an infinite sum of terms calculated from its derivatives at a single point (Maclaurin uses x = 0).

$$f(x) = f(0) + f'(0)x + f''(0)x^2/2! + \ldots$$

Q7. Define Partial Derivatives.

The derivative of a multivariable function with respect to one variable while keeping others constant.

$$\partial f/\partial x$$

Q8. State Euler's Theorem for Homogeneous Functions.

For a homogeneous function f(x, y) of degree n, the sum of variables times their partial derivatives equals n times the function.

$$x(\partial f/\partial x) + y(\partial f/\partial y) = nf$$

Q9. What are Indeterminate Forms?

Expressions like 0/0 or ∞/∞ that do not have a defined value but can be solved using L'Hôpital's rule.

$$\lim [f(x)/g(x)] = \lim [f'(x)/g'(x)]$$

Q10. Explain Lagrange's Multiplier method for extrema.

Used to find local maxima and minima of a function subject to equality constraints.

$$\nabla f = \lambda \nabla g$$

Q11. What is an Indefinite Integral?

The set of all antiderivatives of a function, denoted with a constant of integration C.

$$\int f(x)\, dx = F(x) + C$$

Q12. Define the Definite Integral.

Represents the area under a curve between two points a and b.

$$\int_{a}^{b} f(x)\, dx = F(b) - F(a)$$

Q13. Explain Integration by Substitution.

A method to solve complex integrals by changing the variable of integration.

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

Q14. What is Integration by Parts?

A rule that transforms the integral of a product of functions into a simpler form.

$$\int u\,dv = uv - \int v\,du$$

Q15. Describe Reduction Formulae.

Recursive formulas used to evaluate integrals involving high powers of trigonometric or algebraic functions.

$$I_n = f(I_{n-1}, I_{n-2})$$

Q16. How do you find the Length of a Curve (Arc Length)?

The total distance along a path of a function f(x) from a to b.

$$L = \int_{a}^{b} \sqrt{1 + (f'(x))^2}\,dx$$

Q17. Define Area under a curve.

The space between the function graph and the x-axis, calculated using definite integration.

$$Area = \int f(x)\,dx$$

Q18. What is Volume of Revolution?

The volume of a solid formed by rotating a 2D area around an axis.

$$V = \pi \int [f(x)]^2\,dx$$

Q19. What is a Double Integral?

Integration over a two-dimensional region, used to calculate volume or area in 3D.

$$\iint f(x, y)\,dA$$

Q20. Explain the concept of Maxima and Minima for two variables.

Points where a multivariable function reaches its highest or lowest values in a local region.

$$f_x = 0, f_y = 0$$